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"It's not like I take two pills and then at 10am I take another two. It really depends on what chemicals are brewing in my brain that day."

Michael J. Fox (1)

This observation from Michael J. Fox, founder of *The Michael J. Fox Foundation for Parkinson's Research*, captures one of the fundamental challenges in Parkinson's disease management - unpredictable fluctuations in symptom severity and medication efficacy. This unpredictability, along with symptoms such as bradykinesia, dyskinesia, sleep disturbances, and neuropsychiatric issues (2) are hallmarks of the condition affecting approximately 18,000 individuals in Ireland (3). Although Ireland ranked 'worst for digital health policies in the developed world' (4) in 2023, a shift towards a connected health model is underway, with the implementation of SláinteCare, a comprehensive strategy aimed at improving healthcare delivery, across the country, including an 'eHealth' programme (5). *DopaTrack*, the mobile application idea I helped develop, offers a promising solution to the unpredictable fluctuations Fox describes. However, it is important to acknowledge that Parkinson's presents a wide range of complex challenges, each of which creates a unique opportunity for innovation in digital health. From early detection to advanced stage management, digital technologies have the potential to address diverse aspects of the disease, which could ultimately improve quality of life and provide a continuous model of disease management for individuals like Michael J. Fox who are currently living with Parkinson's.

Disease Background

Parkinson's Disease is characterised by protein aggregation and gradual degeneration of dopaminergic neurons in the substantia nigra (SN) of the midbrain (6). The disease progresses through distinct stages, corresponding to dopamine neuron loss: prodromal, early motor, mid, and advanced stages (7). By the time classical motor symptoms appear (bradykinesia, rigidity, tremor, and postural instability) and a PD diagnosis is made, 50%-60% of dopamine neurons have already been lost (7,8). A key manifestation of bradykinesia in Parkinson's is the development of dysarthria, or speech difficulties, which are seen in approximately 90% of patients and significantly interfere with patient's ability to communicate (9). Recent advancements in speech analysis offer promising opportunities for digital health innovations using speech patterns in Parkinson's (10).

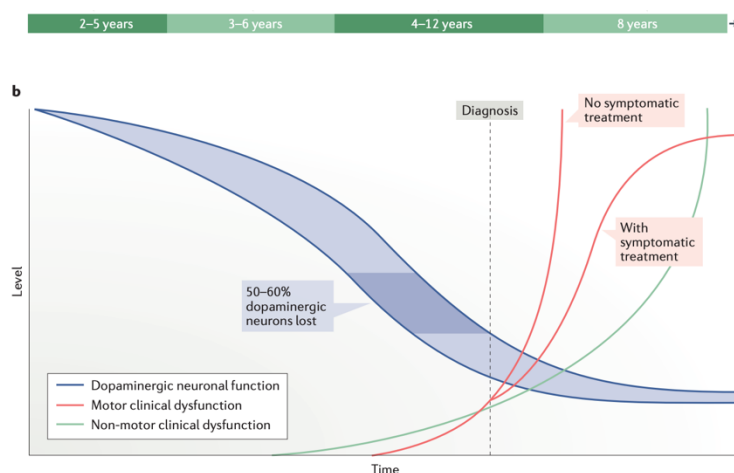


Figure 1: Disease Life Cycle of Parkinson's Disease: Time course of motor and non-motor symptoms (7).

In terms of demographics, PD generally affects the older population (average onset 60 years (11)), with male predominance that has been shown to increase with age (12). The older age profile of the disease presents unique challenges for digital health technology adoption, with approximately 40% of patients experiencing mild cognitive impairment in early disease stages, progressing to dementia in up to 80% of patients in advanced stages (13).

Parkinson's Lifecycle: Opportunities for Change

The Parkinson's disease lifecycle encompasses three key stages of clinical care: pre-diagnosis/early detection, diagnosis and ongoing disease management. In Ireland, PD patients experience a fragmented pathway of care across this cycle; from initial GP visits for non-specific PD symptoms, through lengthy referral wait times, to delayed diagnosis and infrequent follow ups (14). This substandard level of care is partly the result of specialist shortages around the country, with clinical nurse specialist staffing at only 44% of the required level (14). The first national audit on Parkinson's care in 2024 identified many shortcomings across the patient clinical pathway in the Irish health system (15,16). These systemic failures across the PD care cycle are creating opportunities for digital health solutions to transform the current fragmented approach into a continuous, connected care model that extends beyond clinical settings, into patient homes.

“ There is probably nothing like a disaster or pandemic to force the health service to get creative ... I really hope that they don't forget about us on the outskirts though ... It would be a great thing in my opinion if we could stay connected more to the professionals, using technology.

”

Figure 2: Quote from Irish PD patient: Highlighting need for digital health technology (16)

Early Detection

The prodromal stage in Parkinson's is characterised by the appearance of non-motor symptoms such as depression, REM sleep disorders and constipation, as well as minor speech impairments which are often too subtle and non-specific to allow early diagnosis using currently available methods (17). These symptoms may precede the appearance of classic motor symptoms by a decade (18). In the national PD audit, the diagnostic pathway for patients was found to be frequently prolonged. Specifically, it showed that early-onset Parkinson's patients faced discriminatory practices leading to more complicated diagnostic pathways, often hindered by limited GP knowledge in primary care settings (16). These issues demonstrate the need for objective, digital biomarkers to enhance early detection of Parkinson's, in place of the current use of unstandardised, subjective GP assessments.

Digital approaches for detecting prodromal PD remain largely experimental, with limited validated technology in clinical use that can successfully identify the disease before obvious motor symptoms appear (19). However, newly developed algorithms such as PREDICT-PD and PREDIGT are attempting to provide patients with early PD clinical diagnosis using risk assessment strategies. The PREDICT-PD algorithm uses markers such as keyboard finger-tapping speed and probable REM sleep behaviour disorder (RBD) as an estimation of risk (20), while PREDIGT uses a complex structure of interacting contributions from five risk categories (E,D,I,G,T) to provide risk assessments (21). A key limitation of the use of these multi-factor

algorithms is that they assume independent effects for each factor on PD disease risk, which is very difficult to validate and has led to delays in clinical implementation.

Notably, emerging research indicates that speech biomarkers could serve as highly sensitive early disease indicators for PD, as it has been shown that speech impairments precede the appearance of classic motor symptoms by several years (22). In a model for prodromal PD, speech biomarkers were used to classify the disease with 89% accuracy in males and 70% accuracy in females (23), with spontaneous and emotionally charged speech tasks offering the greatest separation between groups. The integration of standalone, objective biomarkers such as speech into the early disease cycle could offer unprecedented opportunities for the diagnosis of individuals; particularly for those whose symptoms were previously believed to be too non-specific to allow for early identification.

Features	Male		Female		P-value	
	PD	HC	PD	HC	Disease effect	Sex effect
Prosody						
SD log F0 (10^{-3} Hz)	58 ± 10	73 ± 13	71 ± 14	76 ± 12	< 0.0001	< 0.0001
SD Int (dB)	5.4 ± 0.5	5.5 ± 0.4	5.3 ± 0.4	5.3 ± 0.4	0.56	0.21
Pauses						
No. pauses (pauses/s)	0.63 ± 0.25	0.65 ± 0.20	0.54 ± 0.19	0.65 ± 0.35	0.17	0.33
No. short pauses (pauses/s)	0.24 ± 0.15	0.30 ± 0.14	0.26 ± 0.13	0.34 ± 0.15	0.001	0.33
Median duration (s)	0.68 ± 0.18	0.56 ± 0.12	0.56 ± 0.13	0.49 ± 0.09	< 0.0001	< 0.0001
Rhythm						
RSD	0.10 ± 0.07	0.07 ± 0.04	0.11 ± 0.09	0.09 ± 0.05	0.003	0.24
Absolute (s)	0.18 ± 0.12	0.14 ± 0.09	0.20 ± 0.11	0.15 ± 0.09	0.017	0.66
Pace (syll/s)	1.02 ± 0.21	0.98 ± 0.18	1.03 ± 0.16	0.96 ± 0.12	0.08	0.97
Phonation						
NHR (%)	4.5 ± 6.9	3.4 ± 3.5	4.0 ± 8.1	3.5 ± 7.8	0.44	0.90
Jitter (%)	1.0 ± 1.2	0.8 ± 0.5	0.9 ± 1.2	0.8 ± 1.0	0.44	0.90
Shimmer (%)	5.5 ± 3.0	5.4 ± 2.1	4.0 ± 2.7	3.6 ± 2.2	0.44	< 0.0001

Means and standard deviations of vocal features related to Prosody, Pauses, Rhythm and Phonation were computed using all the tasks performed by PD and HC participants, recorded with the professional microphone. Statistical differences were performed using two-way analyses of variance with disease status and sex as factors. P-values were adjusted according to false discovery rate correction. Abbreviations: F0: fundamental frequency; HC: healthy control; Int: Intensity; NHR: noise-to-harmonics ratio; PD: Parkinson's disease; RSD: relative standard deviation; SD: standard deviation.

Figure 3: Speech biomarkers table: Table of vocal feature biomarkers investigated for use in early diagnosis of prodromal Parkinson's

Diagnosis

Diagnosis, the second stage of the patient cycle, is the stage that has usually coincided with the appearance of classical motor symptoms in the patient (18). Usually, a diagnosis of PD is made based on patient history and evidence of the cardinal symptom, bradykinesia (with rest tremor and/or rigidity) which is assessed by the clinician through the observation of manual tasks such as finger tapping, foot tapping or hand movements (24,25). The 2024 national audit identified significant disparities between assessments in PD diagnosis across the country as a major barrier to diagnosis for patients and reports showed that over half of PD patients were waiting at least 4 months for a diagnosis (15,16). Inconsistencies around current patient diagnoses presents an opportunity for the application of digital health tools, wearables and apps to clinical pipelines, which could increase the speed and accuracy of diagnosis, given that continuously recorded objective patient metrics can be accessed through their use (26).

Symptoms of Parkinson's that may be quantified using digital tools include serial reaction time (SRT), sleep disturbances, speech disturbances, ocular dysfunction and impaired gait and balance, as well as many others. Numerous digital tools already exist that facilitate this analysis, including QDG-Care system, Dreem2 EEG headband, AHTD speech test and Personal KinetiGraph (PKG) which could provide the data needed to accelerate the route to diagnosis

for many patients (26,27). Notably, speech disturbance analyses have even shown efficacy in distinguishing between PD subtype at diagnosis (22,28). Clinical validation and standardisation of digital tools for their safe, integrated use of in the clinical route to diagnosis is needed to allow the adoption of technologies which will improve patient outcomes (29). These digital diagnostic tools would not only enable standardisation of the diagnosis process but would create a path for continuity across the care pathway into the patient's long term management plan.

Disease Management

The third part of the cycle; disease management and monitoring, is largely characterised by the challenge of managing symptoms associated with medication efficacy fluctuations in PD. The gold standard treatment is levodopa combination treatment, which is classically taken as an oral formulation, or can be taken as an extended-release oral formulation, intestinal gel, or inhalation powder (30). Due to the short half-life and inconsistent absorption pattern of the drug in the intestinal tract, its effects are usually unpredictable, leading to the experience of symptom fluctuations by patients as previously described (30). 'Off' refers to periods when the effects of the medication wears off and motor symptoms of Parkinson's reappear or worsen in the patient, while 'on' periods represent times when medication is well-controlling these symptoms. The 'on/off' phenomenon is described like a 'light switch' for many patients due to how quickly the switch to 'off' and the recurrence of symptoms can occur (31), with switches becoming more severe later in the Parkinson's lifecycle as PD stage advances (32).

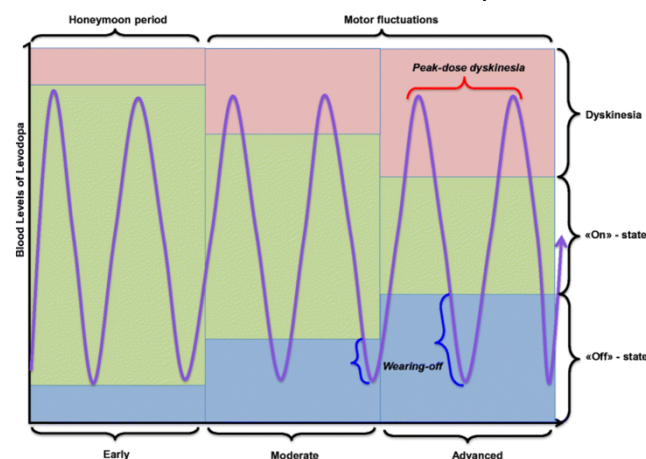


Figure 4: *Parkinson's Disease Lifecycle, 'on/off' Levodopa fluctuations (33)*

From extensive research carried out as part of the design challenge, my team and I identified these fluctuations as a fundamental issue, particularly given the current management pathway that exists in Ireland. Currently, the only recommendation given from Parkinson's Ireland to patients to deal with unpredictable fluctuations is for patients to 'take notes' of the times and frequency of their symptoms and their medication to present to medical staff at reviews, which 43% of patients receive only annually (16,31). Although apps such as StrivePD and MyTherapy already exist which integrate passive data storage from wearables with substantial levels of self-reported symptom/mood data to better track PD, this approach is still partially problematic (34,35). This is due to the fact patients often exhibit altered disease perception, and their self-reported motor symptoms significantly misalign with clinically identified 'on/off' periods, which leads to a substantial gap between perceived and actual symptom states in record taking and disease tracking (36).

In response to these shortfalls, my team and I developed an idea for the integration of semantic and acoustic biomarkers into an app-based platform called *DopaTrack* to help patients continuously better identify and manage their ON/OFF periods. Studies have already

proven the efficacy of several language biomarkers in identifying ON/OFF states with up to 85%-95% accuracy depending on specific speech tasks performed by patients, with highest accuracy achieved using semi-spontaneous speech tasks such as storytelling (37). Tools used to carry out this classification include feature extraction and deep-learning techniques (37,38), which could be seamlessly integrated into an app user-interface to identify states in real-time. The idea is that users could speak into the app at a given point within a medication fluctuation and be provided with an accurate prediction as to the extent to which they were in a delayed on, wearing off, dyskinetic or in complete on/off state. Insights into patients' medication fluctuations using speech biomarkers would provide an objective marker of disease state, enabling the development of personalised medication plans and would allow clinicians to monitor disease progression using app-based data storage, which is especially crucial in advanced disease stages with more severe and frequent fluctuations.

Future Prospects

A fully integrated connected health model for Parkinson's disease could transform the current fragmented care cycle in Ireland into a continuous, data-driven pathway that could solve issues in a similar way to how *DopaTrack* could address the key challenge of Parkinson's medication fluctuations that Fox described (1).

While numerous digital biomarkers show potential for use, speech analysis serves as a key illustrative example of how just a single digital measure could help provide continuity across the entire disease lifecycle. In the prodromal stage, remote speech analysis could enable non-invasive population-level screening, identifying subtle vocal changes years before motor symptoms appear. During diagnosis, these same speech patterns could provide objective confirmation of clinical observations and PD subtype, standardising the diagnostic process. Throughout the management phase, continuous speech monitoring (as demonstrated by *DopaTrack*) would enable real-time assessment of medication efficacy and disease progression through speech biomarker analysis (22). This approach benefits all stakeholders: patients receive continuous monitoring rather than infrequent follow-ups, earlier intervention, and greater autonomy; clinicians receive objective evidence supporting diagnosis and treatment decisions; and healthcare systems reduce emergency interventions while optimising resource allocation. Implementation barriers to widespread adoption in Ireland include privacy concerns, technology accessibility for the older demographic, and integration challenges with Ireland's existing health infrastructure, which has yet to fully transition to an EHR (39).

The Irish context, with its documented resource shortages and fragmented care pathway, presents an ideal opportunity for digital transformation. While Parkinson's provides an example for the potential benefits of this approach, the shift to a connected health model would ideally transform healthcare delivery across all conditions, shifting from reactive, episodic sick-care toward a proactive, continuous, and personalised wellness-based system.

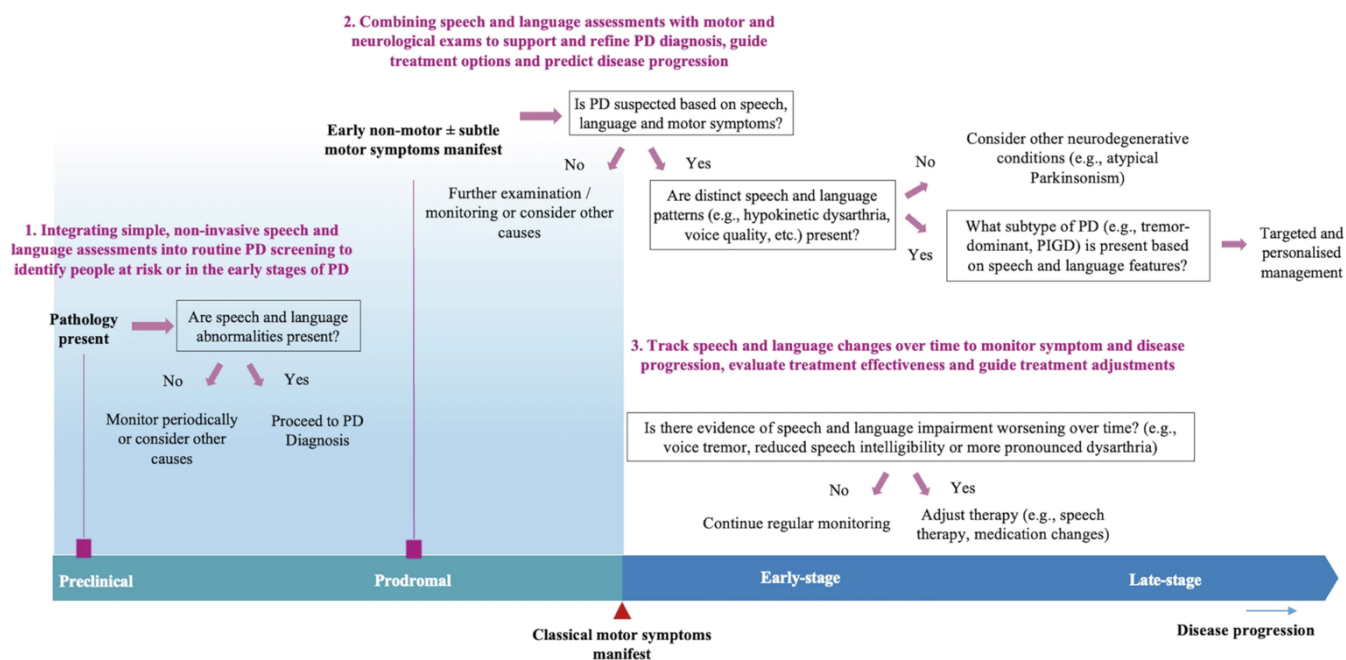


Figure 5: Map of future augmented patient journey for PD using speech biomarkers: An illustrative model of how digital biomarkers and technology can provide the opportunity to integrate the Parkinson's disease lifecycle into a continuous, personalised model of care (22).

Bibliography

1. parkinsonseurope.org. Michael J Fox on living with Parkinson's disease. Stories.
2. Tolosa E, Garrido A, Scholz SW, Poewe W. Challenges in the diagnosis of Parkinson's disease. *Lancet Neurol*. 2021 May;20(5):385–97.
3. Parkinson's Ireland. What is Parkinson's. <https://www.parkinsons.ie//understanding-parkinsons/what-is-parkinsons/>. 2025;
4. Paul Cullen. Ireland ranks worst for digital health policies in developed world, says report. *The Irish Times*. 2023 Nov;
5. Department of Health. Sláintecare Implementation Strategy and Action Plan 2021-2023. 2021 May.
6. Cerri S, Mus L, Blandini F. Parkinson's Disease in Women and Men: What's the Difference? *J Parkinsons Dis*. 2019;9(3):501–15.
7. Schapira AH V, Chaudhuri KR, Jenner P. Non-motor features of Parkinson disease. *Nat Rev Neurosci*. 2017 Jul;18(7):435–50.
8. Kalia L V, Lang AE. Parkinson's disease. *Lancet*. 2015 Aug 29;386(9996):896–912.
9. Atalar MS, Oguz O, Genc G. Hypokinetic Dysarthria in Parkinson's Disease: A Narrative Review. *Sisli Etfal Hastan Tip Bul*. 2023;57(2):163–70.
10. Ngo QC, Motin MA, Pah ND, Drotár P, Kempster P, Kumar D. Computerized analysis of speech and voice for Parkinson's disease: A systematic review. *Comput Methods Programs Biomed*. 2022 Nov;226:107133.
11. National Institute on Aging (NIH). NIH (NIH.gov). 2022. Parkinson's Disease: Causes, Symptoms, and Treatments.
12. Moisan F, Kab S, Mohamed F, Canonico M, Le Guern M, Quintin C, et al. Parkinson disease male-to-female ratios increase with age: French nationwide study and meta-analysis. *J Neurol Neurosurg Psychiatry*. 2016 Sep;87(9):952–7.
13. Fang C, Lv L, Mao S, Dong H, Liu B. Cognition Deficits in Parkinson's Disease: Mechanisms and Treatment. *Parkinsons Dis*. 2020;2020:2076942.
14. O' Shea E, Rukundo A, Foley G, Wilkinson T, Timmons S. Experiences of health service access: A qualitative interview study of people living with Parkinson's disease in Ireland. *Health Expectations*. 2024 Feb 5;27(1).
15. UCC PINN. AUDIT OF MOVEMENT DISORDER SERVICES IN THE REPUBLIC OF IRELAND. 2024 Apr.
16. O' Shea E, Rukundo A, Bennett K, Foley G, Wilkinson T, Timmons S. Experiences of Health Service Access and Use for People Living with Parkinson's Disease in Ireland: A National Survey. *Health Soc Care Community*. 2023 Aug 9;2023:1–9.
17. Jackson H, Anzures-Cabrera J, Simuni T, Postuma RB, Marek K, Pagano G. Identifying prodromal symptoms at high specificity for Parkinson's disease. *Front Aging Neurosci*. 2023 Sep 22;15.
18. Tolosa E, Garrido A, Scholz SW, Poewe W. Challenges in the diagnosis of Parkinson's disease. *Lancet Neurol*. 2021 May;20(5):385–97.
19. Serag I, Azzam AY, Hassan AK, Diab RA, Diab M, Hefnawy MT, et al. Multimodal diagnostic tools and advanced data models for detection of prodromal Parkinson's disease: a scoping review. *BMC Med Imaging*. 2025 Mar 28;25(1):103.
20. Bestwick JP, Auger SD, Simonet C, Rees RN, Rack D, Jitlal M, et al. Improving estimation of Parkinson's disease risk—the enhanced PREDICT-PD algorithm. *NPJ Parkinsons Dis*. 2021 Apr 1;7(1):33.
21. Li J, Mestre TA, Mollenhauer B, Frasier M, Tomlinson JJ, Trenkwalder C, et al. Evaluation of the PREDIGT score's performance in identifying newly diagnosed Parkinson's patients without motor examination. *NPJ Parkinsons Dis*. 2022 Jul 29;8(1):94.

22. Cao F, Vogel AP, Gharahkhani P, Renteria ME. Speech and language biomarkers for Parkinson's disease prediction, early diagnosis and progression. *NPJ Parkinsons Dis.* 2025 Mar 24;11(1):57.
23. Jeancolas L, Mangone G, Petrovska-Delacrétaz D, Benali H, Benkelfat BE, Arnulf I, et al. Voice characteristics from isolated rapid eye movement sleep behavior disorder to early Parkinson's disease. *Parkinsonism Relat Disord.* 2022 Feb;95:86–91.
24. Kobylecki C. Update on the diagnosis and management of Parkinson's disease. *Clin Med (Lond).* 2020 Jul;20(4):393–8.
25. Postuma RB, Berg D, Stern M, Poewe W, Olanow CW, Oertel W, et al. MDS clinical diagnostic criteria for Parkinson's disease. *Mov Disord.* 2015 Oct;30(12):1591–601.
26. Sun Y meng, Wang Z yun, Liang Y yuan, Hao C wei, Shi C he. Digital biomarkers for precision diagnosis and monitoring in Parkinson's disease. *NPJ Digit Med.* 2024 Aug 21;7(1):218.
27. Hoffman SL, Schmiedmayer P, Gala AS, Wilkins KB, Parisi L, Karjagi S, et al. Comprehensive real time remote monitoring for Parkinson's disease using Quantitative DigitoGraphy. *NPJ Parkinsons Dis.* 2024 Jul 27;10(1):137.
28. Rusz J, Krupička R, Vítěčková S, Tykalová T, Novotný M, Novák J, et al. Speech and gait abnormalities in motor subtypes of de-novo Parkinson's disease. *CNS Neurosci Ther.* 2023 Aug;29(8):2101–10.
29. Coravos A, Khozin S, Mandl KD. Developing and adopting safe and effective digital biomarkers to improve patient outcomes. *NPJ Digit Med.* 2019 Mar 11;2(1):14.
30. Urso D, Chaudhuri KR, Qamar MA, Jenner P. Improving the Delivery of Levodopa in Parkinson's Disease: A Review of Approved and Emerging Therapies. *CNS Drugs.* 2020 Nov;34(11):1149–63.
31. Parkinson's Ireland. On/Off Motor Fluctuation in Parkinson's Disease. <https://www.parkinsons.ie/understanding-parkinsons/symptoms/movement-related-symptoms/on-off-in-parkinsons/>. 2025;
32. Aquino CC, Fox SH. Clinical spectrum of levodopa-induced complications. *Movement Disorders.* 2015 Jan 8;30(1):80–9.
33. You H, Mariani LL, Mangone G, Le Febvre de Nailly D, Charbonnier-Beaupel F, Corvol JC. Molecular basis of dopamine replacement therapy and its side effects in Parkinson's disease. *Cell Tissue Res.* 2018 Jul 7;373(1):111–35.
34. parkinsons.org.uk/catalogue. Tech Guide. 2025. StrivePD.
35. parkinsons.org.uk/catalogue. Tech Guide. MyTherapy.
36. Löhle M, Bremer A, Gandor F, Timpka J, Odin P, Ebersbach G, et al. Validation of the PD home diary for assessment of motor fluctuations in advanced Parkinson's disease. *NPJ Parkinsons Dis.* 2022 Jun 2;8(1):69.
37. Pompili A, Solera-Ureña R, Abad A, Cardoso R, Guimarães I, Fabbri M, et al. Assessment of Parkinson's Disease Medication State Through Automatic Speech Analysis. In: *Interspeech 2020*. ISCA: ISCA; 2020. p. 4591–5.
38. Norel R, Agurto C, Rice JJ, Ho BK, Cecchi GA. Speech-based identification of L-DOPA ON/OFF state in Parkinson's Disease subjects. 2018.
39. Reynolds J. An integrated approach to digital health. Medical Independent (medicalindependent.ie). 2024 Aug 26;